

5.13.2

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Question

Find the equation of the circle having $(1, -2)$ as its centre and passing through the intersection of

$$3x + y = 14, \quad 2x + 5y = 18. \quad (1)$$

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

$$\begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 14 \\ 18 \end{pmatrix} \quad (2)$$

$$\mathbf{B} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \quad (3)$$

circle equation is

$$\mathbf{x}\mathbf{x}^T + 2\mathbf{u}^T\mathbf{x} + f = 0 \quad (4)$$

$$\mathbf{u} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (5)$$

Theoretical Solution

point **B** lies on circle,so

$$\mathbf{b}\mathbf{b}^T + 2\mathbf{u}^T\mathbf{b} + f = 0 \quad (6)$$

$$f = -20 \quad (7)$$

hence the required circle equation is

$$\mathbf{x}\mathbf{x}^T + 2 \begin{pmatrix} 1 & -2 \end{pmatrix} \mathbf{x} - 20 = 0 \quad (8)$$

```
#include <stdio.h>
#include <math.h>

int main() {
    // Circle centre
    double h = 1, k = -2;

    // Solve intersection of  $3x + y = 14$  and  $2x + 5y = 18$ 
    // Equation 1:  $3x + y = 14 \rightarrow y = 14 - 3x$ 
    // Substitute in  $2x + 5y = 18$ 
    double x, y;
    x = (18 - 5*14) / (2 - 15); // Simplified from substitution
    y = 14 - 3*x;

    // Radius = distance between (h,k) and intersection point
    double r = sqrt(pow(x - h, 2) + pow(y - k, 2));
}
```

```
// Print results
printf("Intersection Point: (%.2f, %.2f)\n", x, y);
printf("Centre: (%.2f, %.2f)\n", h, k);
printf("Radius: %.2f\n", r);
printf("Equation of Circle: (x - %.2f)^2 + (y - %.2f)^2 = %.2f^2\n", h, k, r);

return 0;
}
```

```
#Code by GVV Sharma (modified for circle through line
    intersection)
#Sep 27, 2025
#Released under GNU GPL

import sys
import numpy as np
import matplotlib.pyplot as plt
from numpy import linalg as LA

# Local imports from CoordGeo
sys.path.insert(0, '/sdcard/github/matgeo/codes/CoordGeo')
from line.funcs import line_isect, line_norm
from conics.funcs import circ_gen

# Line equations:  $3x + y = 14$ ,  $2x + 5y = 18$ 
n1 = np.array([3,1]).reshape(-1,1)
c1 = 14
```

```
n2 = np.array([2,5]).reshape(-1,1)
c2 = 18

# Intersection of lines
P = line_isect(n1,c1,n2,c2)

# Circle centre
O = np.array([1,-2]).reshape(-1,1)

# Radius = distance between centre and intersection point
r = LA.norm(O-P)

# Generate circle
x_circ = circ_gen(O,r)

# Generate lines
x_line1 = line_norm(n1,c1,-5,5)
x_line2 = line_norm(n2,c2,-5,5)
```



```
# Plotting
plt.plot(x_circ[0,:], x_circ[1,:], label="Circle")
plt.plot(x_line1[0,:], x_line1[1,:], label="$3x+y=14$")
plt.plot(x_line2[0,:], x_line2[1,:], label="$2x+5y=18$")

# Points
plt.scatter(P[0],P[1],color='red')
plt.scatter(O[0],O[1],color='blue')
plt.annotate("Intersection P", (P[0],P[1]), textcoords="offset
points", xytext=(10,10))
plt.annotate("Centre O", (O[0],O[1]), textcoords="offset points",
xytext=(10,-10))
```

```
# Axes formatting
ax = plt.gca()
ax.spines['top'].set_color('none')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
ax.spines['bottom'].set_position('zero')
plt.legend()
plt.grid()
plt.axis("equal")

# Save and show
plt.savefig("circle_question.png", dpi=300)
plt.show()
```

Plot

